

Effort Estimation of Agile Development using Fuzzy Logic

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Abstract: Effort Estimation is one of the most important factors for a successful project. Mostly it is done in the starting of the project life cycle. The vital activity in Agile project organization is a prediction of Software development this is because a project that is available in the beginning is not sufficient and moreover it is not clear because of the confliction. Getting accurate Effort Estimation for any software project has been a challenge always. In this paper fuzzy logic model is used to find Effort Estimation of Agile software development. In this model Fuzzy method is used to take the raw data of actual figures and output is given as the estimated effort available from industry. This way is more suitable when data is not available for analyzing by orthodox method. Basic idea is taken from Human where action and Inaction are not obvious, they are not clear. In this paper we will use three input variable to calculate Efforts estimation i.e User story, Team Expertise and Complexity. In this manner, it is quite difficult to discover goals in more exact and clear simple way.

Keywords: User story, Effort Estimation, Agile Software Development, Fuzzy logic

I. INTRODUCTION

Agile Development is a term that consists of many different parts of iterative and incremental software development methods. It promotes modifying arrangements with minimum time, evolutionary improvement and delivery, a time-based iterative approach that is enough flexible and responsive to change. It is a planned structure that advances predicted connections all through the improvement cycle. Each of the Agile methods is one of a kind in its particular approach; they all offer a typical vision and common goal of their project. It is vital to incorporate iteration and regular comments that it gives to consecutively refine and bring a software system. It consists of consistent scheduling, testing from the first phase, regular integration, and other forms of development of both the undertaking project and the software. They are generally unimportant, particularly contrasted with orthodox waterfall-style processes also called sequential process. What is more critical about Agile techniques is that they all pay attention to engaging persons to work together and settle on choices that are more effective for the goal well defined in advance [2].

The exactness in the effort estimation is extremely important in light of the fact that both underestimation and overestimation are unsafe to the organizations. In the project, if estimation is too low at a point then there will be a pressure to finish the task at the earliest opportunity. On opposite side, if the effort is too high, at that point number of resources will be required. It

suggests that effort estimation is a difficult problem and it is vital to examine a fitting technique for development of estimation accuracy. The introduction summarized that effort estimation in Agile development is the basic action to be performed and to take care of the issues in regards to effort estimation that could save more time and cost of the project. It is recommended to use fuzzy logic as it is better to solve the present challenge of effort estimation.

A. Agile versus Traditional Method

If a right method for software development is chosen, it can help to save the extra cost and improve the quality of the deadline. Conventional methods get the job done when project depend on inflexible output to yield quality. In search technique, we can't make changes in between the phases if we want to until the entire phase of the cycle is complete [10]. In this world of Technology, the dynamic change of project and need should always be fulfilled [12].

The ancient method of software development includes more hard phases. These phases have well set of approach for requirement gathering. This way we can easily find the gap between the actual and desired result as this is the main purpose of a unit and integration testing. This helps to make flawless quality principles that help in further development of the project.

B. Characteristics of Agile Software Process

- Iterative: Agile software forms center around short cycles. Inside each cycle, a specific arrangement of activities is finished to accomplish the goal that is defined in an early stage.
- Versatile: The alert procedure adjusts the procedure to find newly discovered risk, uncovered in any cycle. Similarly Agile process suit any additional movement or change to the current activity
- Incremental: An Agile procedure does not try to construct the whole framework without a moment's delay.
- Joined: The fundamental premise of an Agile process is to construct the framework nearer to the fact. This objective is accomplished by applying every single conceivable system to guarantee achievement in the quickest form.
- Individuals Oriented: Agile procedures support individuals completed procedure and innovation. They develop through adaptation in a natural way. Developers help to

increase the efficiency, quality and the time required for execution of result. They are the best folks in the association to know how to give improvements.

- **Collective:** Agile procedures nurture correspondence among coworkers. Rapidly incorporating an extensive task while additions are being produced in parallel, requires a coordinated effort.

C. Agile Method

Agile methods are more concerned about the customer needs and few ambitions. These Method can be adjusted with device autonomously[13], [14]. This method helps to increase overall flexibility. They focus more on people and connections and in addition consumer Meeting rather than phases, following an inflexible arrangement.

II. RESEARCH PROBLEM

The vast amount of present effort estimation measures is created in order to help customer consecutive development philosophies though Agile Software Development is iterative in nature. On the off chance that these regular frameworks are used for exertion estimation of Agile software ventures, by then the precision of result will be low. The present routine with respect to exertion estimation for Agile software ventures relies upon single cycle. In this way, an effort estimation show is relied upon to see advancement effort of Agile software venture, in perspective of characteristics of Agile Software Development approach [5]. The significance and trouble of project arranging and particularly effort estimation fluctuates by groups, organizations, and projects. Contrasted with Agile arranging, where groups typically plan the next short sprint, arranging a whole Waterfall SDLC process can be a challenging work [6], [9].

Gathering and sorting out recorded information, and analyzing it to enhance the precision of appraisals for future projects must be proficiently performed by utilizing software tools that help these procedures.

III. PROPOSED MODEL

In this paper, MATLAB is used to calculate the effort estimation in Agile development using Fuzzy logic. MATLAB stands for matrix laboratory. MATLAB is a user-friendly structure whose central data part acts a cluster [4], [17].

Means a does not require any size[15]. This helps us to overcome various problems of processing, especially with mathematical design, in minimum time it came give a solution to the complex structure and input. In industry, this tool is used for high-gainfulness research, progression, and examination with accumulated prerequisites.

A. Fuzzy Logic

Fuzzy logic is an effective strategy to tackle the practical issues which incorporate the effort less ness and adaptability. It handles the issues with loose and inadequate information

which depends on the fuzzy set hypothesis based on limited information. The benefit of using range is that the software developers can easily predict the effort a project that has a different model. A system based on FL has a direct relationship with fuzzy concepts and fuzzy logic.

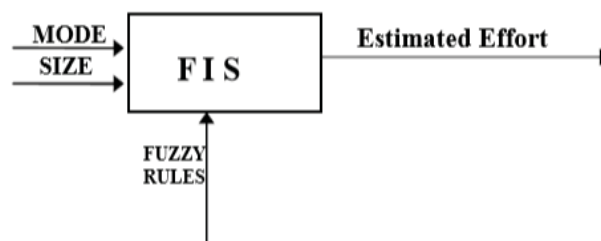


Fig. 1. Fuzzy Structure

Fuzzy Framework or structure is shown in fig.1. The input to this structure is a user story, team expertise, and complexity. There are different fuzzy ranges which are used to take input from a different variable. We have collected project metrics datasheet from different industries where we have observed the total number of defects before the delivery of the project, Project goals, total defects found and actual effort. The fuzzy method is used to identify the size of project size instead of any specific number[1]. We will use Min_Max Formula to calculate the output effort using Fuzzy logic structure(Max-Min/Max-v)

The model used in MATLAB is exposed in Fig.2

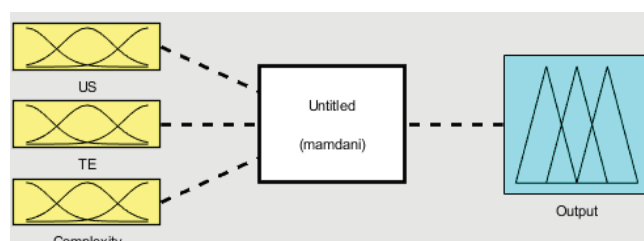


Fig. 2. Fuzzy Process converts the input values from the user and gives output with the help of its processing part

The popular fuzzy logic systems can be categorized into three types: Pure fuzzy logic systems, Takagi and Sugeno's fuzzy system, and fuzzy logic system with fuzzifier and defuzzifier. Since most of the engineering applications produce crisp data as input and expects crisp data as output, the last type is the most widely used type of Fuzzy logic systems[16]. A fuzzy logic system with fuzzifier and defuzzifier, first, proposed by Mamdani and it has been successfully applied to a variety of industrial processes and consumer products [7], [8].

IV. EXPERIMENTAL STUDY

In order to measure estimation accuracy of the Agile model, we collected data previously developed projects from 6 software houses. These projects have been developed using

Agile Software development methodology. This experimental analysis was performed by a group of research students, who were unaware of actual results.

In carrying out our experiments, we have used 3 variables Low (L), Medium (M) and High (H) as shown in the screenshots below

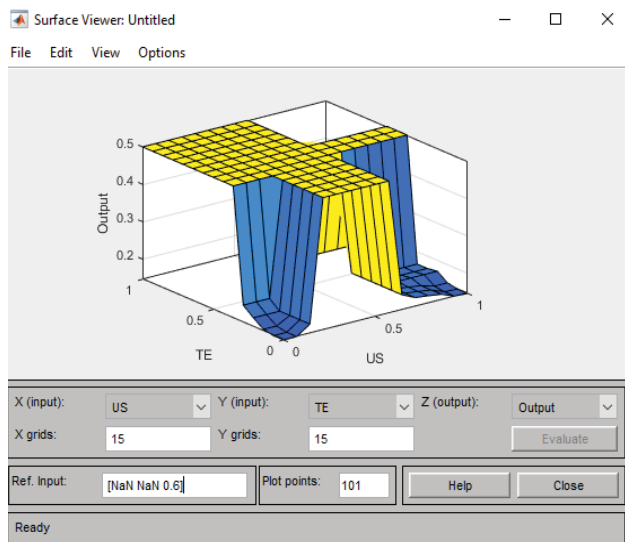


Fig. 3. Surface Viewer

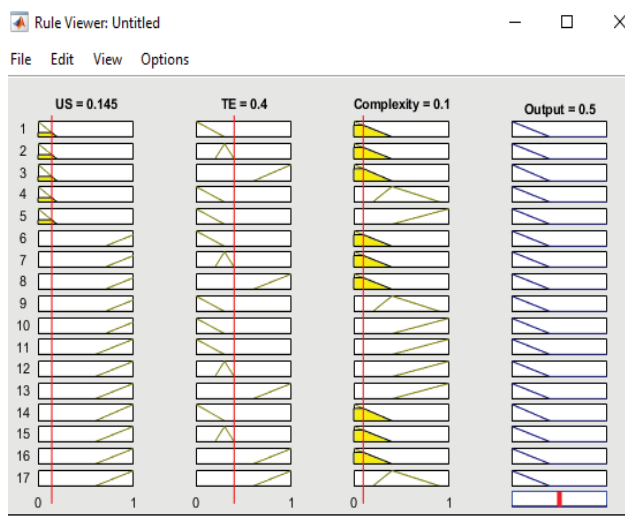


Fig. 4. Rule Viewer Output

We have considered 3 input variable to calculate Efforts estimation

1) *User Story*: If there are more story points then effort estimation will be more. The exact effort depends upon the variables we are using with the User story. The measure of work expected to execute a single functionality of the item that is being built[20]. This measure is not Specifically connected to the execution time of the user story.. It is critical for the

complexities of the user stories from the reference set to be defined as early as possible. The next user stories are estimated by comparing them with this set, with the nature of further estimations being dependent on this accuracy. A reference client story should, thus, be picked deliberately

The effort of a specific A Story Point is the measure of effort, finished in a unit time.

2) *Team Expertise*: It is one of the important factors in deciding the time and quality factor in Agile development. There is the huge difference in the outcome of efforts if the whole team has enough experience to save the time and cost of software development. Story points are a theoretical measure and thus they may cause a few troubles when a new group chooses to utilize them

3) *Complexity*: This consist of either or both the desires of the Story and or its practical difficulty. Complexity presents the risk to the estimate; more difficulty means more amounts of ambiguity and increase in the risk factor[18]. Like Story Size, these guidelines are not fixed. These can be adjusted by the team itself; however, we have categorized them to accommodate all characteristics of Agile software development methodology.

A. Fuzzy Rules

Our rule-based on User stories (US), Team Expertise (TE) and Complexity (C) appears in the following form

If the US is low, TE is low and C is also low then output effort will also be high

If the US is high, TE is medium and C is low then output effort will be Medium

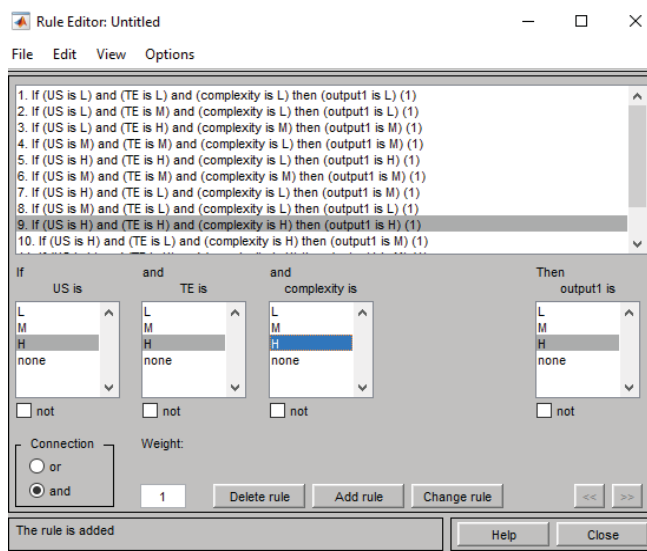


Fig. 5. Rule Editor

V. ESTIMATION OF DELIVERY DATE

To evaluate conveyance the information chosen needs should be executed in the present phase. The phase of planning depends on the velocity and in the requirement gathering changes as per the input need of the user. The requirement gathering consists of the client stories that need to be evaluated[21].

The estimated delivery date mostly depend on the expertise of the team, if the team will have more experience then they will take less time and can deliver the project within the deadline with proper quality of work[3]. If the same project is done by the new team then it will be going to take more time. After each Story is part of assignments the gathering picks regardless of whether they can center around passing on that Story amid the Iteration. The estimate should be taken in such a way, even if there is a small change and the requirement the team won't cross the deadline.

VI. CONCLUSION

This paper presents work on Effort Estimation using Fuzzy Logic Model with MATLAB. There is a chance of improvement in the process of taking input from the user that is called Fuzzyfication. If this process of Fuzzyfication gets better then it will produce a better result in future and the input required by this process engine will also be low this will also help to consume lower time, ultimately increasing the productivity of software development. The model uses User Stories, Team Expertise, and complexity as the base for estimation. We have to use Fuzzy Logic just because we were having limited dataset available from the industry. The effort estimation of any project completely relies on the input variable we are considering so it is very important to select those input parameters selectively. In the initial phase of any software development the accurate and dependable estimate is a little bit difficult task for software managers when the software estimation are judged by humans, the results of the estimation are quite different than that when we have user input data available from the industry. The main purpose of an Agile Software Development is to find solutions for the problem of which we could have available resources within that gives result in deadline. This type of software development model has replanning at different stages so it is easy for us to find solutions at different phases of software development and testing it all together. User stories just give an estimated input required by the developer. It will not be 100% accurate but it can reduce the risk of uncertainty which could lead to lots of time and resources wastage.

VII.FUTURE WORK

Effort estimation can also find out using the neural network in future but a neural network is beneficial only if we have a large amount of dataset. There is a lot of upcoming work that can be done using neuro-fuzzy approach. It allows us to use the numerical data and team expertise all together so that best

result could find out and the chance of risk and failure will also reduce

Evaluation based on User story, Team Expertise and Complexity help us to conclude that Story size will also help to know the estimate of work we need to do in order to make the project successful with increased efficiency. It will not be 100% accurate but it can reduce the risk of uncertainty which could lead to lots of time and resources wastage.

ACKNOWLEDGEMENT

Authors express their deep sense of gratitude to The Founder President of Amity University, Dr. Ashok K. Chauhan for his keen interest in promoting research in the Amity University and have always been an inspiration for achieving great heights.

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